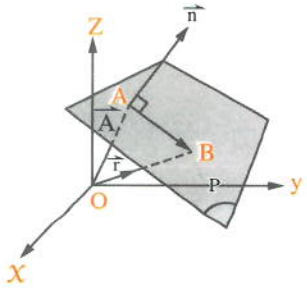
Unit 4

Lesson Two

Equation of a plane in space

Different forms of the equation of the plane in space

Let A (x_1 , y_1 , z_1) be a given point on the plane P its position vector $\vec{A}$ and $\vec{n} = (a , b , c)$ be the normal direction vector to the plane "Remember that all the perpendicular vectors to the plane are parallel" , let B (x , y , z) be any point on the plane P its position vector $\vec{r}$, then



$\because \vec{n} \perp$ plane P $\quad\quad \therefore \vec{n} \perp$ any vector in the plane

$\therefore \vec{n} \perp \overrightarrow{AB} \quad\quad \therefore \vec{n} . \overrightarrow{AB} = 0$

$\therefore \vec{n} . \left(\vec{B} - \vec{A} \right) = 0 \quad , \quad \because \vec{B} = \vec{r}$

$\therefore \vec{n} . \left(\vec{r} - \vec{a} \right) = 0 \ (1) \quad\quad \therefore \vec{n} . \vec{r} - \vec{n} . \vec{A} = 0$

$\therefore \vec{n} . \vec{r} = \vec{n} . \vec{A}$ "vector form of the equation of the plane"

, from (1) :

, $\because \vec{n} = (a , b , c) , \vec{r} = (x , y , z) , \vec{A} = (x_1 , y_1 , z_1)$

$\therefore (a , b , c) . (x - x_1 , y - y_1 , z - z_1) = 0$

$\therefore a (x - x_1) + b (y - y_1) + c (z - z_1) = 0$ "The standard form of the equation of the plane"

, by expanding

$\therefore a x + b y + c z + (- a x_1 - b y_1 - c z_1) = 0$

, putting $- a x_1 - b y_1 - c z_1 = d$

$\therefore a x + b y + c z + d = 0$ "The general form of the equation of the plane"

Notice that

$d = - \vec{n} . \vec{A}$

from the previous it is clear that the plane P is determine by a point on it and a normal direction vector to it.

- **From our previous study , we know that the plane is determined also by :**

1. Three non-collinear points lie on it.

2. Two intersecting straight lines lie on it.

3. A straight line and a point does not belong to it lie on the plane.

4. Two different parallel straight lines.

Example 1

Find the different forms of the equation of the plane passing through the point $(2 , -3 , 5)$ and the direction $\vec{n} = (-1 , 2 , 3)$ is normal direction vector on it.

Solution

- Vector form :

$\because \vec{n} . \vec{r} = \vec{n} . \vec{A}$

$\therefore (-1 , 2 , 3) . \vec{r} = (-1 , 2 , 3) . (2 , -3 , 5)$

$\therefore (-1 , 2 , 3) . \vec{r} = 7$

- Standard form :

$\because a (x - x_1) + b (y - y_1) + c (z - z_1) = 0$

$\therefore -1 (x - 2) + 2 (y + 3) + 3 (z - 5) = 0$

- General form : $-x + 2y + 3z - 7 = 0$

Example 2

Find the different forms of the equation of the plane passing through the points A $(1 , 3 , 5)$, B $(-1 , 2 , 4)$ and C $(4 , 4 , 0)$

Solution

$\overrightarrow{AB} = \vec{B} - \vec{A} = (-2 , -1 , -1)$, $\overrightarrow{AC} = \vec{C} - \vec{A} = (3 , 1 , -5)$

, $\because \frac{-2}{3} \neq \frac{-1}{1}$ $\therefore \overrightarrow{AB}$ is not parallel to $\overrightarrow{AC}$

$\therefore$ A , B and C are three non collinear points.

- To find the equation of the plane we need :

1. a given point on it "There are three given points"

2. a normal direction vector to the plane and we can get it from $\overrightarrow{AB} \times \overrightarrow{AC}$

$$\therefore \vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -1 & -1 \\ 3 & 1 & -5 \end{vmatrix} = 6\hat{i} - 13\hat{j} + \hat{k}$$

- The vector form of the plane equation : $\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{A}$

  $\therefore (6 , -13 , 1) \cdot \vec{r} = (6 , -13 , 1) \cdot (1 , 3 , 5)$

  $\therefore (6 , -13 , 1) \cdot \vec{r} = -28$

- The standard form : $6(x - 1) - 13(y - 3) + (z - 5) = 0$
- The general form : $6x - 13y + z + 28 = 0$

Remark

Equation of the plane passing through the three non collinear points (x_1 , y_1 , z_1) , (x_2 , y_2 , z_2) and (x_3 , y_3 , z_3) is :

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

- It is possible to use the previous remark to find the equation of the plane in the previous example as follows :

$$\begin{vmatrix} x - 1 & y - 3 & z - 5 \\ -1 - 1 & 2 - 3 & 4 - 5 \\ 4 - 1 & 4 - 3 & 0 - 5 \end{vmatrix} = 0 \qquad \therefore \begin{vmatrix} x - 1 & y - 3 & z - 5 \\ -2 & -1 & -1 \\ 3 & 1 & -5 \end{vmatrix} = 0$$

- The standard form of the plane equation : $6(x - 1) - 13(y - 3) + (z - 5) = 0$
- The general form : $6x - 13y + z + 28 = 0$
- The vector form : $(6 , -13 , 1) \cdot \vec{r} = -28$

Notice that

- Components of the normal direction vector of the plane are the coefficients of x , y , z in the general solution of the plane.
- $\vec{n} \cdot \vec{A}$ in the vector equation of the plane $= -d$ in the general equation.

Remarks

1. Two straight lines in space : as solving two straight lines equations simultaneous and :

(1) The S.S. = $\varnothing$, then the two lines are skew or parallel

(2) The S.S. = one point , then the two lines are intersecting and contained in one plane

- If the two lines intersect in more than one point , then they are coincident.

2. Straight line and plane in space : as solving the equations of a straight line and a plane simultaneously and :

(1) The S.S. = $\varnothing$, then the straight line is parallel to the plane

(2) The S.S. = one point , then the straight line intersects the plane in this point

- If the straight line and the plane are intersecting in more than one point , then the plane contains this line.

Example 3

Prove that : the two straight lines $L_1 : \overrightarrow{r_1} = (3 , -2 , -1) + t_1 (2 , 3 , 1)$, $L_2 : \overrightarrow{r_2} = (4 , 5 , 1) + t_2 (1 , 7 , 2)$ are intersecting then find the point of intersection and equation of the plane containing them.

Solution

The two straight lines intersect at $\overrightarrow{r_1} = \overrightarrow{r_2}$

i.e. $(3 , -2 , -1) + t_1 (2 , 3 , 1) = (4 , 5 , 1) + t_2 (1 , 7 , 2)$

, $\because$ the corresponding coefficients of the two sides are equal

$\therefore 3 + 2 t_1 = 4 + t_2$ and hence $2 t_1 - t_2 = 1$ (1)

, $-2 + 3 t_1 = 5 + 7 t_2$ and hence $3 t_1 - 7 t_2 = 7$ (2)

, $-1 + t_1 = 1 + 2 t_2$ and hence $t_1 - 2 t_2 = 2$ (3)

Solving the two equations (1) , (3) :

$\therefore t_1 =$ zero , $t_2 = -1$

, substituting in (2) we find that $t_1 =$ zero , $t_2 = -1$ satisfying it.

$\therefore$ The two straight lines are intersecting at $t_1 =$ zero in equation of L_1

and $t_2 = -1$ in equation of L_2

, from equation of L_1 :

$\therefore$ Point of intersection is $(X , y , z) = (3 , -2 , -1) +$ zero $(2 , 3 , 1)$

i.e. The intersection point = $(3 , -2 , -1)$

, $\because \overrightarrow{d_1}$ (direction vector of L_1) = (2 , 3 , 1)

, $\overrightarrow{d_2}$ (direction vector of L_2) = (1 , 7 , 2)

$\therefore \overrightarrow{n}$ (normal direction vector to the plane) = $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 7 & 2 \end{vmatrix} = -\hat{i} - 3\hat{j} + 11\hat{k}$

, taking any point lie on L_1 or L_2

$\therefore$ The vector equation of the plane : $(-1 , -3 , 11) . \overrightarrow{r} = (-1 , -3 , 11) . (3 , -2 , -1)$

i.e. $(-1 , -3 , 11) . \overrightarrow{r} = -8$

, the standard form : $-(x-3) - 3(y+2) + 11(z+1)$

, the general form : $-x - 3y + 11z + 8 = 0$

Example 4

Prove that : the two straight lines $L_1 : \overrightarrow{r} = (2 , 3 , -4) + t(2 , 3 , 1)$

, $L_2 : \frac{x-5}{4} = \frac{y+4}{6} = \frac{z-4}{2}$ are parallel , then find the general equation of the plane contains these two lines.

Solution

$\because \overrightarrow{d_1} = (2 , 3 , 1)$, $\overrightarrow{d_2} = (4 , 6 , 2)$

, $\because \frac{2}{4} = \frac{3}{6} = \frac{1}{2}$

$\therefore$ The two lines are parallel or coincident

, $\because$ the point A $(2 , 3 , -4) \in L_1$, and by substitution in L_2 it does not satisfy its equation

$\therefore$ The two lines are parallel and not coincident

$\therefore$ There is a unique plane contains them

, let A $(2 , 3 , -4) \in L_1$, B $(5 , -4 , 4) \in L_2$

$\therefore \overrightarrow{AB} = B - A = (3 , -7 , 8)$

$\therefore$ The vector direction perpendiculer to the required plane

$= \overrightarrow{AB} \times \overrightarrow{d_1} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -7 & 8 \\ 2 & 3 & 1 \end{vmatrix} = -31\hat{i} + 13\hat{j} + 23\hat{k}$

$\therefore$ The equation of the plane is $(-31 , 13 , 23) . \overrightarrow{r} = (-31 , 13 , 23) . (2 , 3 , -4)$

i.e. $31x - 13y - 23z = 115$

Another solution

Choose three different non collinear points , that can identify the plane A $(2 , 3 , -4) \in d_1$

, B $(5 , -4 , 4) \in L_2$, C $(4 , 6 , -3) \in L_1$ (by taking $t = 1$ in L_1)

$\therefore$ The plane equation is $\begin{vmatrix} x-2 & y-3 & z+4 \\ 5-2 & -4-3 & 4+4 \\ 4-2 & 6-3 & -3+4 \end{vmatrix}$ = zero

By expanding the determinant $\therefore 31x - 13y - 23z = 115$

Example 5

Find the intersection point of the straight line $2x = 3y - 1 = z - 4$ with the plane : $3x + y - 2z = 5$

Solution

From the equation of the plane :

$y = 5 + 2z - 3x$, substituting in the straight line equation

$\therefore 2x = 3(5 + 2z - 3x) - 1 = z - 4$

$\therefore 2x = 15 + 6z - 9x - 1 = z - 4$ $\therefore 2x = 14 + 6z - 9x = z - 4$

$\therefore 2x = 14 + 6z - 9x$ $\therefore 11x - 6z = 14$ (1)

, $14 + 6z - 9x = z - 4$ $\therefore 9x - 5z = 18$ (2)

Solving (1) , (2) : $\therefore x = -38$, $z = -72$

, substituting in the plane equation

$\therefore y = -25$ $\therefore$ intersection point is $(-38 , -25 , -72)$

Another solution

From the straight line equation : $2x = 3y - 1 = z - 4 = t$

$\therefore x = \frac{t}{2}$, $y = \frac{t+1}{3}$, $z = t + 4$

, substituting in the plane equation

, $3\left(\frac{t}{2}\right) + \frac{t+1}{3} - 2(t + 4) = 5$ (multiplying $\times 6$)

$\therefore 9t + 2t + 2 - 12t - 48 = 30$ $\therefore t = -76$

$\therefore x = \frac{-67}{2} = -38$, $y = \frac{-76+1}{3} = -25$, $z = -76 + 4 = -72$

$\therefore$ The intersections point is $(-38 , -25 , -72)$

Notice that

If t has only one value then the straight line intersects the plane.

Example 6

Prove that the straight line $\vec{r} = (2, 3, -8) + t(-3, 12, 2)$ is parallel to the plane $2x + y - 3z = 5$

Solution

$\because$ The parametric equations of the straight line are

$x = 2 - 3t$, $y = 3 + 12t$, $z = -8 + 2t$

By substitution in the plane equation

$\therefore 2(2 - 3t) + (3 + 12t) - 3(-8 + 2t) = 5$

$\therefore 4 - 6t + 3 + 12t + 24 - 6t = 5$ $\quad\therefore 31 = 5$ (impossible)

$\therefore$ There is no value of t that makes a point on the straight line lies on the plane

$\therefore$ The straight line parallel to the plane.

Notice that

If "t has no value" then the straight line parallel to the plane.

Example 7

Prove that the straight line $\vec{r} = (1, 2, 3) + t(3, 1, -14)$ lies in the plane $4x + 2y + z = 11$

Solution

$\because$ The parametric equations of the straight line $x = 1 + 3t$, $y = 2 + t$, $z = 3 - 14t$,

By substitution in the plane equation : $\quad\therefore 4(1 + 3t) + 2(2 + t) + (3 - 14t) = 11$

$\therefore 4 + 12t + 4 + 2t + 3 - 14t = 11$

$\therefore 11 = 11$ (Satisfied for all values of t)

$\therefore$ t has infinite number of values

$\therefore$ The straight line lies in the plane.

Notice that

If "t has infinite number of values" Then the straight line lies in the plane.

Another solution

$\because$ The point A $(1, 2, 3) \in$ the straight line

, $\because 4(1) + 2(2) + (3) = 11$ $\quad\therefore$ The point A satisfies the plane equation

put $t = 1$ $\quad\therefore$ The point B $(4, 3, -11) \in$ the straight line

, $\because 4(4) + 2(3) + (-11) = 11$ $\quad\therefore$ The point B satisfies the plane equation

$\therefore$ The two points A , B lie on the straight line and satisfy the plane equation

$\therefore$ The straight line lies in the plane

Equation of the plane knowing the intercepted parts from the coordinate axes :

If the plane cuts the coordinates axes at the points $(x_1 , 0 , 0) , (0 , y_1 , 0) , (0 , 0 , z_1)$, then equation of the plane is :

$$\begin{vmatrix} x - x_1 & y & z \\ -x_1 & y_1 & 0 \\ -x_1 & 0 & z_1 \end{vmatrix} = 0$$

$\therefore y_1 z_1 (x - x_1) - (- x_1 z_1) y + (x_1 y_1) z = 0$

$\therefore y_1 z_1 x - x_1 y_1 z_1 + x_1 z_1 y + x_1 y_1 z = 0$

$\therefore y_1 z_1 x + x_1 z_1 y + x_1 y_1 z = x_1 y_1 z_1$

, dividing by $x_1 y_1 z_1$

$$\therefore \boxed{\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1}$$

"Equation of the plane knowing the intercepted parts from the coordinate axes"

Example 8

Find the equation of the plane passing through the points :

$(0 , 0 , -7)$, $(3 , 0 , 0)$, $(0 , -5 , 0)$

Solution

Equation of the plane is : $\frac{x}{3} + \frac{y}{-5} + \frac{z}{-7} = 1$

Remark

If equation of the plane is : $\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1$

and the plane intersects the coordinate axes at the points A $(x_1 , 0 , 0)$, B $(0 , y_1 , 0)$ and C $(0 , 0 , z_1)$

Then the volume of the pyramid OABC $= \frac{1}{6} | x_1 y_1 z_1|$ cubic units

For example : If the plane $2x + 3y + 4z = 12$

i.e. $\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$ intersects the coordinate axes at the points A , B and C , then volume of the pyramid OABC $= \frac{1}{6} | 6 \times 4 \times 3 | = 12$ cubic units.

Example 9

The plane : $8x + 2y + 4z = 8$ intersects the coordinate axes x , y , z at the points A , B and C respectively , find the area of Δ ABC , then find volume of the pyramid OABC

Solution

Equation of the plane is :

$8x + 2y + 4z = 8$ (dividing by 8) $\quad \therefore \frac{x}{1} + \frac{y}{4} + \frac{z}{2} = 1$

$\therefore$ The plane intersects the coordinate axes at

A $(1 , 0 , 0)$, B $(0 , 4 , 0)$, C $(0 , 0 , 2)$

$\therefore \overrightarrow{AB} = \overrightarrow{B} - \overrightarrow{A} = (-1 , 4 , 0)$, $\overrightarrow{AC} = \overrightarrow{C} - \overrightarrow{A} = (-1 , 0 , 2)$

$\therefore$ Area of Δ ABC $= \frac{1}{2} \| \overrightarrow{AB} \times \overrightarrow{AC} \|$

$\because \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 4 & 0 \\ -1 & 0 & 2 \end{vmatrix} = 8\vec{i} + 2\vec{j} + 4\vec{k}$

$\therefore \| \overrightarrow{AB} \times \overrightarrow{AC} \| = \sqrt{64 + 4 + 16} = 2\sqrt{21}$ $\quad \therefore$ Area of Δ ABC $= \frac{1}{2} \times 2\sqrt{21} = \sqrt{21}$ square units.

, volume of the pyramid OABC $= \frac{1}{6} | 1 \times 4 \times 2 | = \frac{4}{3}$ cubic units

Important Remarks

From the general equation of the plane P : $ax + by + cz + d = 0$, we deduce that :

1. (a , b , c) is normal direction vector to the plane P , $d = -\overrightarrow{A} . \vec{n}$ where $\overrightarrow{A}$ is the position vector of a point $\in$ the plane and $\vec{n}$ is a normal direction vector.
2. (a , b , c) is normal direction vector to any other plane parallel to the plane P
3. If $d = 0$, then the origin point is contained in the plane.
4. If $a = 0$, then the equation is $by + cz + d = 0$ and it is equation of plane parallel to x-axis and perpendicular to yz
   If $b = 0$, then the equation is $ax + cz + d = 0$ and it is equation of plane parallel to y-axis and perpendicular to xz
   If $c = 0$, then the equation is $ax + by + d = 0$ and it is equation of plane parallel to z-axis and perpendicular to xy

For example :

$2x - 3y = 6$ is equation of a plane parallel to z-axis and perpendicular to the plane xy

5. If

$a = d = 0$	then the equation is	$b\,y + c\,z = 0$	and it is equation of plane	contains $\mathcal{X}$-axis	and perpendicular to	$y\,z$
$b = d = 0$		$a\,\mathcal{X} + c\,z = 0$		contains y-axis		$\mathcal{X}\,z$
$c = d = 0$		$a\,\mathcal{X} + b\,y = 0$		contains z-axis		$\mathcal{X}\,y$

For example :

$2\,\mathcal{X} - 3\,y = 0$ is equation of a plane contains z-axis and perpendicular to the plane $\mathcal{X}\,y$

6. The equation of the plane $\mathcal{X}\,y$ is $z = 0$, the equation of the plane parallell to $\mathcal{X}\,y$-plane and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is $z = z_1$

- The equation of the plane $y\,z$ is $\mathcal{X} = 0$, the equation of the plane parallel to $y\,z$ and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is $\mathcal{X} = \mathcal{X}_1$
- The equation of the plane $\mathcal{X}\,z$ is $z = 0$, the equation of the plane parallel to $\mathcal{X}\,z$ and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is $y = y_1$

For example :

$\mathcal{X} = 6$ is the equation of a plane parallel to $y\,z$-plane , $z = -4$ is the equation of a plane parallel to $\mathcal{X}\,y$-plane.

7. Notice the difference between :

- $y + z = a$ is an equation of a plane parallel to $\mathcal{X}$-axis.
- $\mathcal{X} = b$ is equation of a plane parallel to the plane $y\,z$
- $y + z = a$, $\mathcal{X} = b$ is an equation of a straight line. (the intersection line of the previous two planes)

8. If $E(\mathcal{X}_1 , y_1 , z_1)$, $F(\mathcal{X}_2 , y_2 , z_2)$, $G(\mathcal{X}_3 , y_3 , z_3)$ are three points in the space and substituting by each of them in the plane equation gives :

- $a\,\mathcal{X}_1 + b\,y_1 + c\,z_1 + d = 0$
- $a\,\mathcal{X}_2 + b\,y_2 + c\,z_2 + d > 0$
- $a\,\mathcal{X}_3 + b\,y_3 + c\,z_3 + d < 0$

That means $E(\mathcal{X}_1 , y_1 , z_1)$ belonges to plane P and each of $F(\mathcal{X}_2 , y_2 , z_2)$ and $G(\mathcal{X}_3 , y_3 , z_3)$ does not belong to the plane and each of them lies in different side from , the plane P

Example 10

Find the plane equation which is parallel to X-axis and passes through the two points $(1, 2, 3), (-1, 4, 7)$

Solution

$\because$ The plane is parallel to X-axis $\therefore$ The plane equation is $b y + c z + d = 0$

, $\because$ $(1, 2, 3)$ lies in the plane $\therefore$ $2b + 3c + d = 0$ (1)

, $\because$ $(-1, 4, 7)$ lies in the plane $\therefore$ $4b + 7c + d = 0$ (2)

From (1) , (2) : $\therefore$ $2b + 4c = 0$ $\therefore$ $\boxed{b = -2c}$

From (1) : $\therefore$ $-4c + 3c + d = 0$ $\therefore$ $\boxed{d = c}$

$\therefore$ The plane equation is : $-2c y + c z + c = 0$ **i.e.** $-2y + z + 1 = 0$

Example 11

Find the plane equation which contains the y-axis and passes through the point $(2, 1, -3)$

Solution

$\because$ The plane contains the y-axis $\therefore$ The plane equation is $a X + c z = 0$

, $\because$ $(2, 1, -3)$ lies in the plane $\therefore$ $2a - 3c = 0$ $\therefore$ $\boxed{a = \frac{3}{2} c}$

$\therefore$ The plane equation is : $\frac{3}{2} c X + c z = 0$ **i.e.** $3X + 2z = 0$

Example 12

Determine the position of each of the points A $(-3, 3, 5)$, B $(6, 5, 1)$ and C $(6, -5, 1)$ with respect to the plane $2X - 3y + 4z - 5 = 0$

Solution

Substituting by the points in the equation of the plane :

- A $(-3, 3, 5)$: $2(-3) - 3(3) + 4(5) - 5 =$ zero $\therefore$ A $\in$ the plane
- B $(6, 5, 1)$: $2(6) - 3(5) + 4(1) - 5 = -4$ $\therefore$ B $\notin$ the plane
- C $(6, -5, 1)$: $2(6) - 3(-5) + 4(1) - 5 = 26$ $\therefore$ C $\notin$ the plane

$\therefore$ B , C don't belong to the plane and each of them lie in different side from the plane.

Example 13

Find the general equation of the plane (where t_1 , t_2 are parameters) :

$(X, y, z) = (3, 2, 5) + t_1(-3, -3, 4) + t_2(1, 2, -1)$

Solution

$\because$ $(X, y, z) = (3, 2, 5) + t_1(-3, -3, 4) + t_2(1, 2, -1)$

$\therefore$ $X = 3 - 3t_1 + t_2$ (1) , $y = 2 - 3t_1 + 2t_2$ (2) , $z = 5 + 4t_1 - t_2$ (3)

By adding (1) , (3) : $\therefore \mathcal{X} + z = 8 + t_1$ $\quad \therefore t_1 = \mathcal{X} + z - 8$ (4)

, subtracting (1) from (2) : $\therefore y - \mathcal{X} = -1 + t_2$ $\quad \therefore t_2 = y - \mathcal{X} + 1$ (5)

substituting from (4) , (5) in (1) :

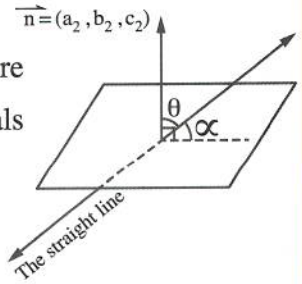
$\therefore \mathcal{X} = 3 - 3(\mathcal{X} + z - 8) + (y - \mathcal{X} + 1)$

$\therefore$ The general equation of the plane is : $5\mathcal{X} - y + 3z - 28 = 0$

The angle between straight line and a plane in space [$\propto$]

If $\vec{d} = (a_1 , b_1 , c_1)$ is direction vector of the straight line and $\vec{n} = (a_2 , b_2 , c_2)$ is normal direction vector to the plane, then measure of the smaller angle between the straight line and the plane ($\propto$) equals (the complements of the angle θ between the two vectors) where

$$\cos\theta = \frac{|(a_1 , b_1 , c_1) . (a_1 , b_2 , c_2)|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\ \sqrt{a_2^2 + b_2^2 + c_2^2}} \quad , \propto = 90° - \theta$$



Example 14

Find the measure of the smaller angle between the straight line :

$\frac{\mathcal{X} - 5}{2} = \frac{y}{-1} = \frac{z + 4}{3}$ and the plane : $3\mathcal{X} + 2y + z - 5 = 0$

Solution

The required angle is the complements of θ where

$$\cos\theta = \frac{|(2 , -1 , 3) . (3 , 2 , 1)|}{\sqrt{4 + 1 + 9}\ \sqrt{9 + 4 + 1}} = \frac{7}{14} = \frac{1}{2} \qquad \therefore \theta = 60°$$

$\therefore$ Measure of the angle between the straight line and the plane $= 90° - 60° = 30°$

Example 15

Find the projection of the point A $(1 , 2 , -1)$ on the plane $2\mathcal{X} + 3y - z + 5 = 0$

Solution

Let the projection of the point A on the plane is the point B $(\mathcal{X}_1 , y_1 , z_1)$

, $\because \overrightarrow{AB}$ // the normal direction vector

$\therefore$ The direction vector of $\overleftrightarrow{AB}$ is $(2 , 3 , -1)$

$\therefore$ The equation of the straight line passing through A and perpendicular to the plane

is $(x , y , z) = (1 , 2 , -1) + t (2 , 3 , -1)$

$\therefore (x , y , z) = (1 + 2t , 2 + 3t , -1 - t)$ (1)

, $\because$ B $\in$ the plane $\therefore$ B satisfies the plane equation

$\therefore 2 (1 + 2t) + 3 (2 + 3t) - (-1 - t) + 5 = 0$

$\therefore 2 + 4t + 6 + 9t + 1 + t + 5 = 0$ $\therefore 14t = -14$ $\therefore t = -1$

by substitution in (1) : $\therefore$ B $= (-1 , -1 , 0)$

Example 16

Find the image of the point A $(1 , 3 , 4)$ by reflection in the plane : $2x - y + z + 3 = 0$

Solution

$\because$ The normal direction vector on the given plane $= (2 , -1 , 1)$

$\therefore$ The equation of the straight line passing through A $(1 , 3 , 4)$ and perpendicular to the plane is $(x , y , z) = (1 + 2t , 3 - t , 4 + t)$

, let C (x_2 , y_2 , z_2) is the image of A by reflection on the plane

$\therefore (x_2 , y_2 , z_2) = (1 + 2t , 3 - t , 4 + t)$ (1)

, $\because$ The midpoint of $\overline{AC}$ is the point B $\in$ the plane

$\therefore$ B satisfies the plane equation

$\therefore 2 \left(\frac{1 + 1 + 2t}{2}\right) - \left(\frac{3 + 3 - t}{2}\right) + \left(\frac{4 + 4 + t}{2}\right) + 3 = 0$

$\therefore 2 + 2t - 3 + \frac{1}{2} t + 4 + \frac{1}{2} t + 3 = 0$ $\therefore 3t = -6$ $\therefore t = -2$

By substitution in (1) : $\therefore$ C $= (x_2 , y_2 , z_2) = (-3 , 5 , 2)$